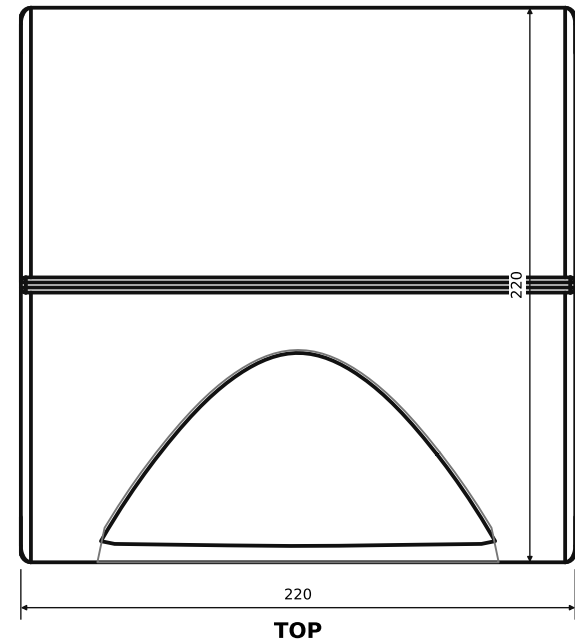
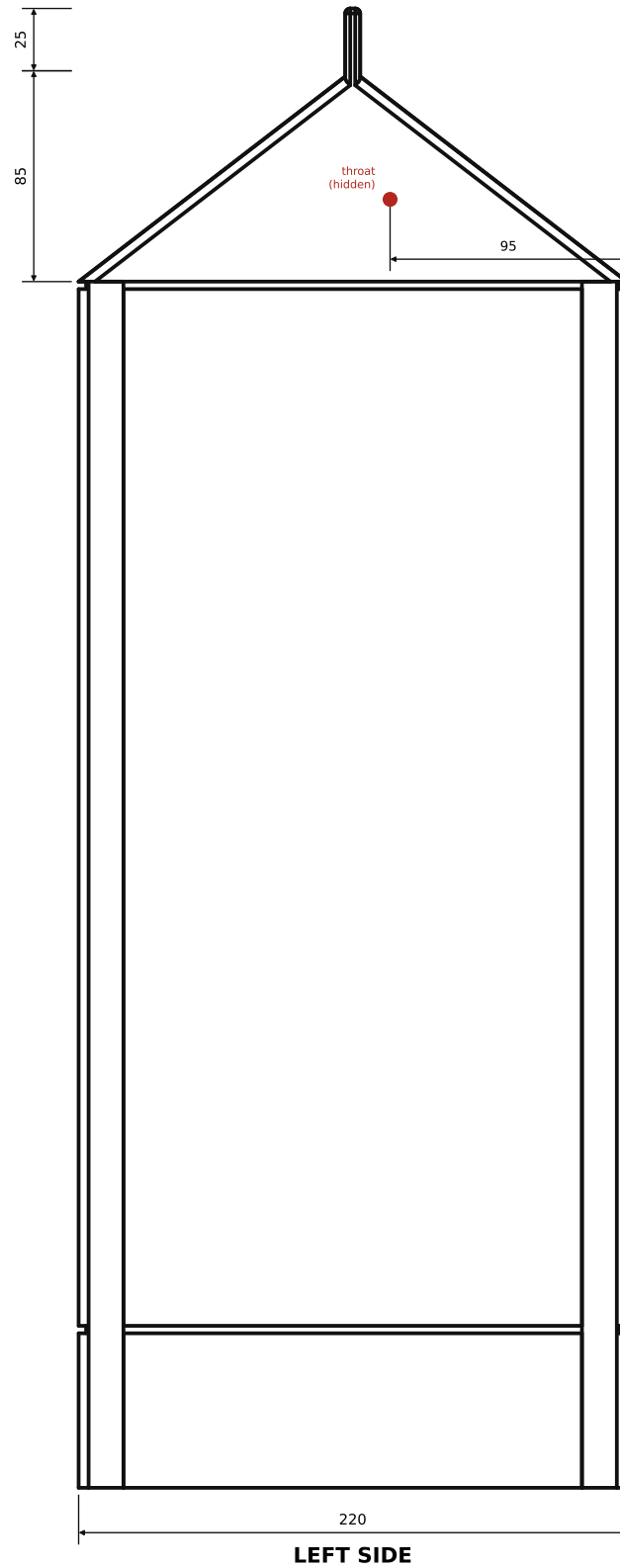
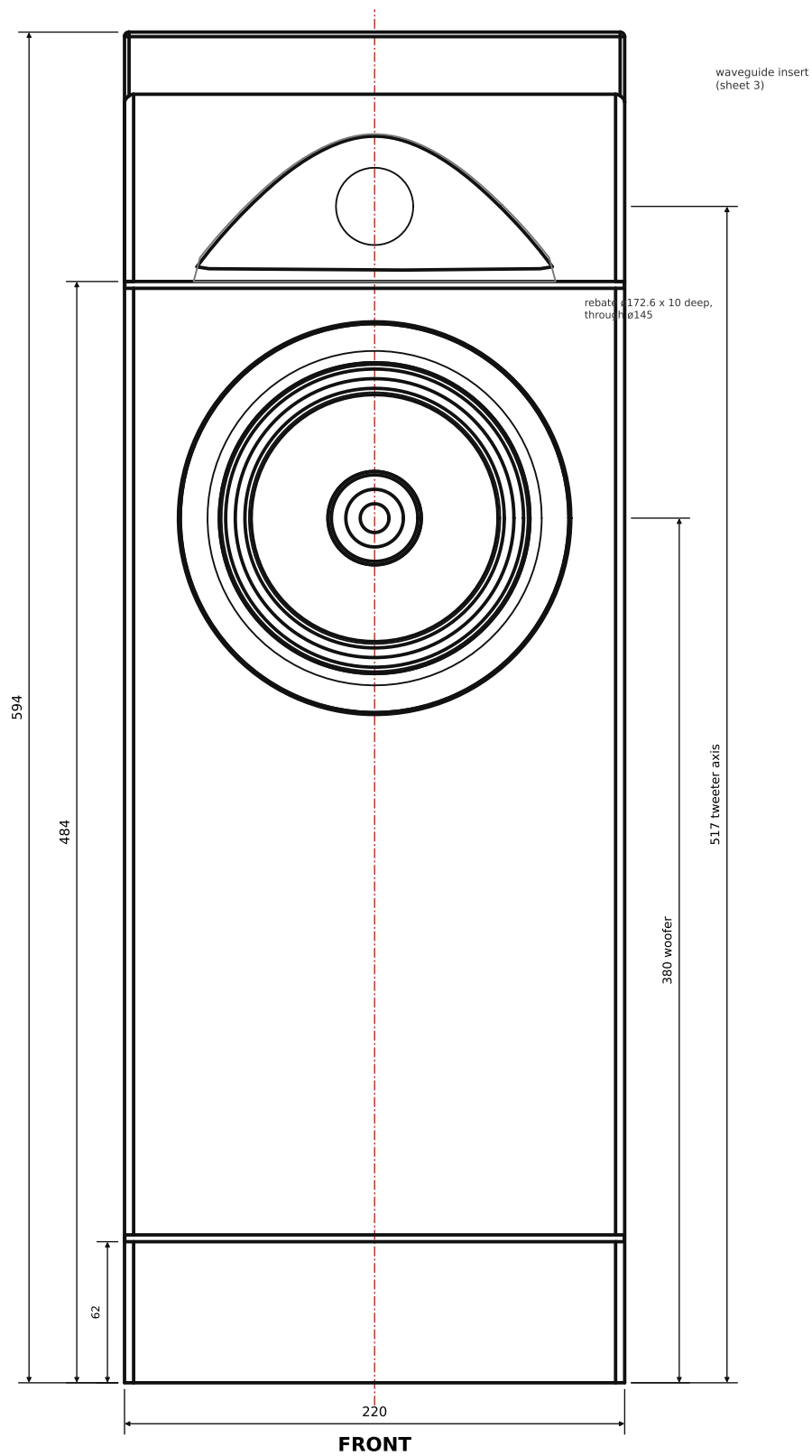




black: the waveguide's mouth, where its lip meets the roof
grey: the insert's seam

Notes

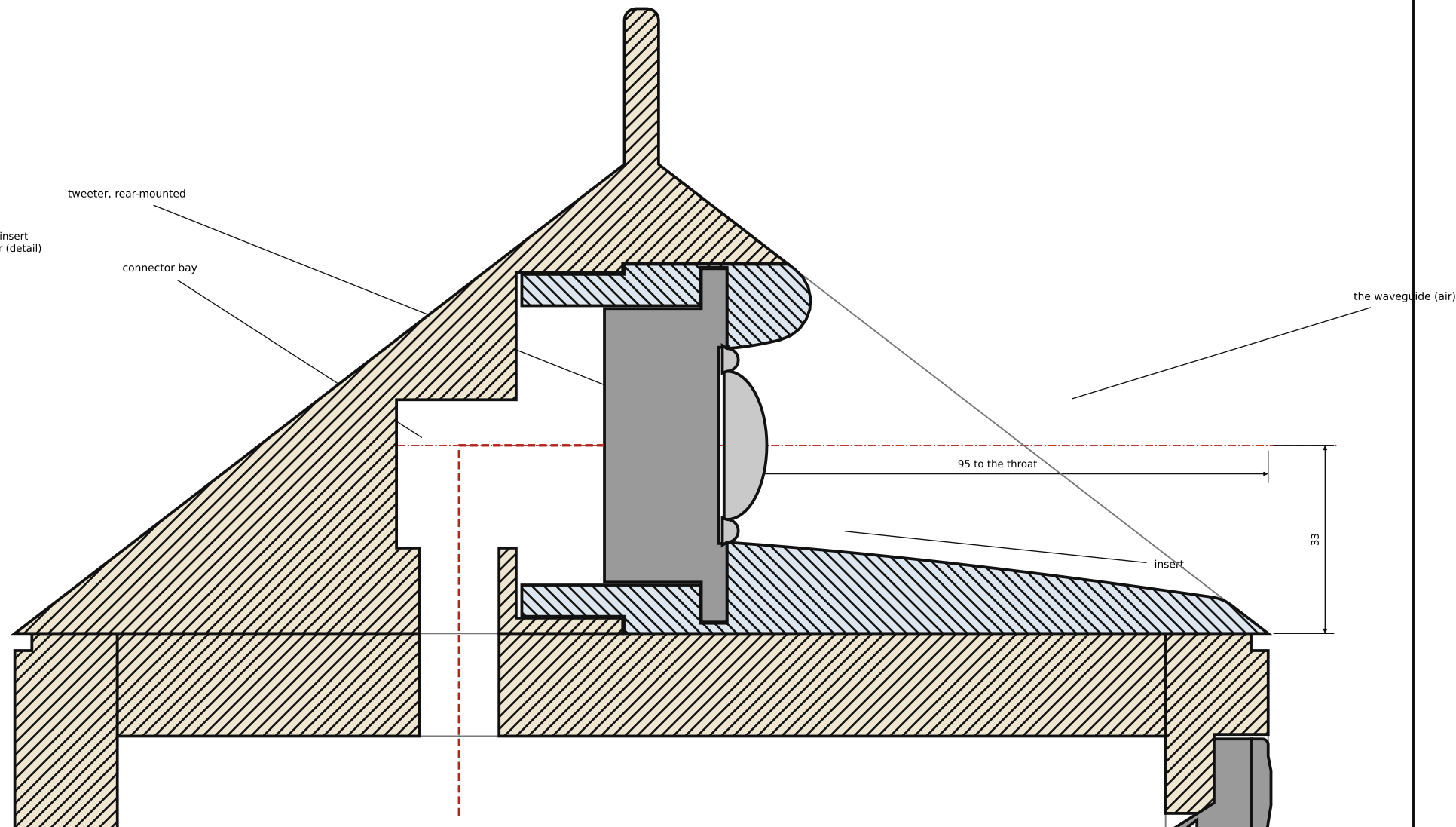
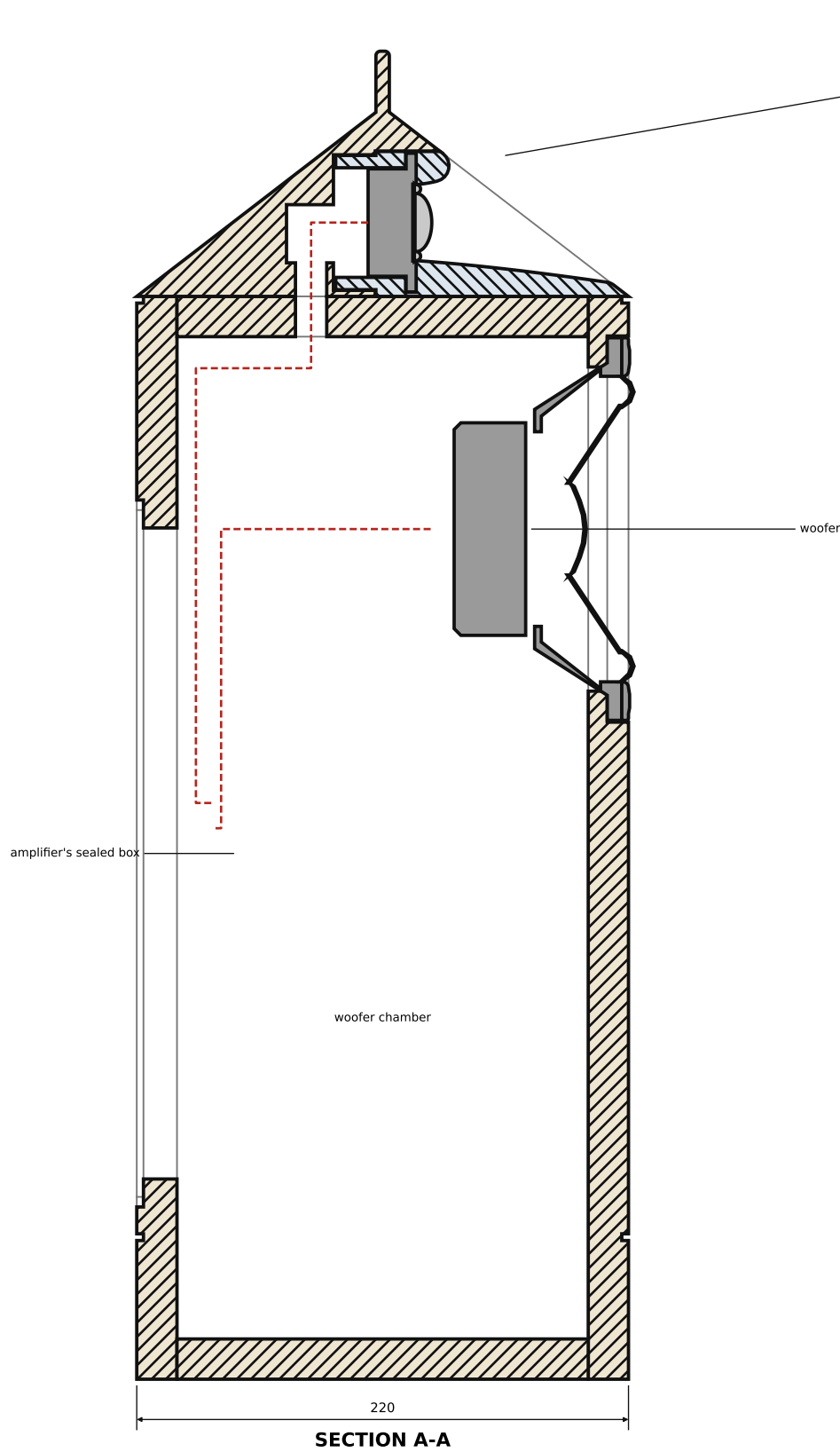
1. Cabinet: 18 mm Baltic birch, glued. The vertical corners and the gable's hips are rounded 4 mm, the fin's edges 2 mm. 3 x 3 mm shadow lines at z 62 and z 481.
2. The drivers sit flush: each frame in a rebate, a 3 mm trim ring over its flange and screws, level with the finish.
3. The tweeter is at the throat of a waveguide insert in the roof (sheet 3): axis z 517, 95 behind the front face. The shape is set by simulation (fab/out/acoustics/waveguide).
4. The back (the amplifier) is drawn on sheet 4 with the wiring; the Nutrition Facts are printed on the right side.
5. Finish: colour coat and 2K clear; the Facts printed between them.

earmilk bookshelf | General arrangement

Front, left side and top at 1:3. Heights from the floor; depths from the front face.

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Check every bought part against its own drawing before cutting.



The roof

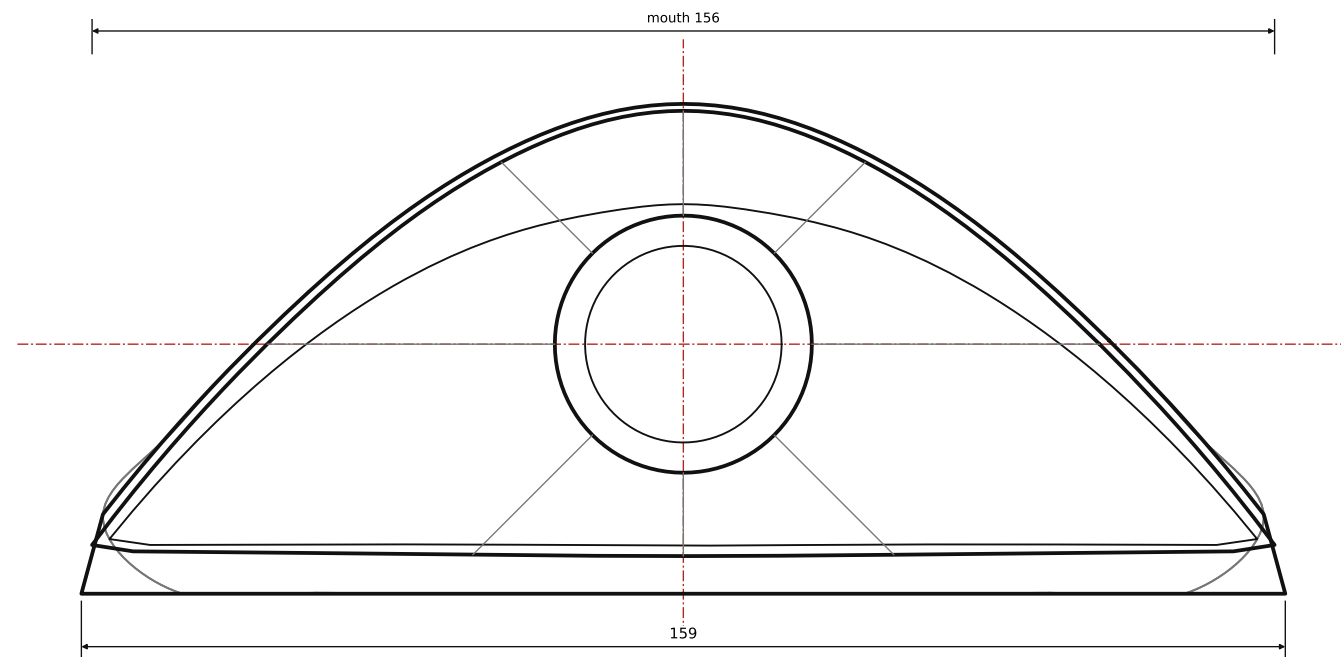
1. The waveguide insert (light blue) sits in a pocket in the gable block, 0.3 mm clear all round. It slides out forward, level, with the tweeter on it.
2. The tweeter is rear-mounted: its flange in a counterbore behind the throat, its dome and surround filling the 34 mm throat, so the wall runs on from the surround with no step.
3. Its lead leaves the boss's open back into the connector bay behind it ($\varnothing 26 \times 22$ deep), where it plugs into the cabinet's lead; that runs down a 14 mm channel through the block and the top panel, sealed by a grommet under the top panel.
4. Four magnets and two pins in the insert's back face meet their partners in the pocket's back wall (sheet 3).

earmilk bookshelf | Section on the centreline

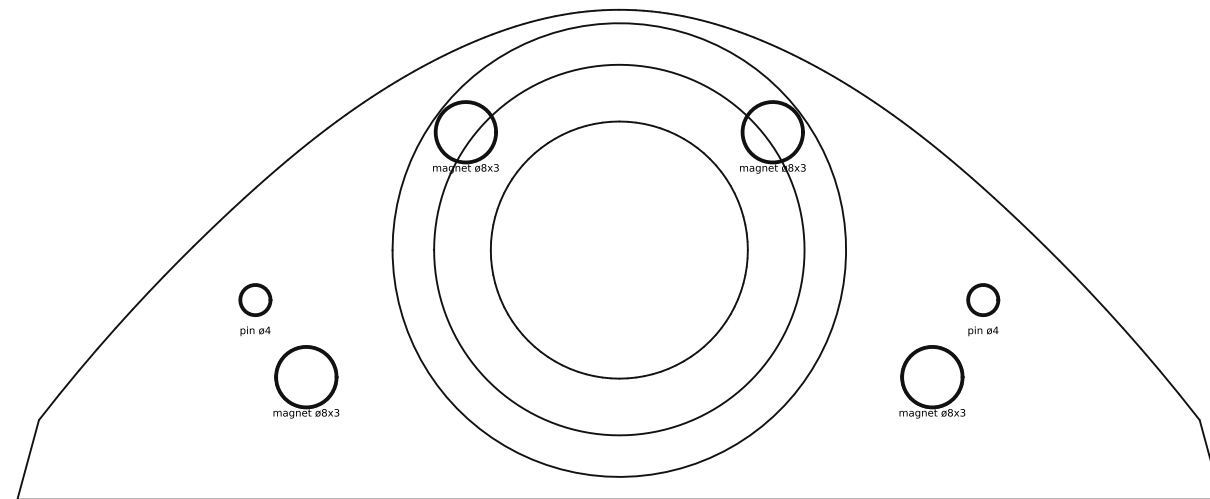
Section A-A on the centre plane (x 110) seen from the left at 1:3, and the roof at 1:1. Cut wood hatched; the insert hatched the other way.

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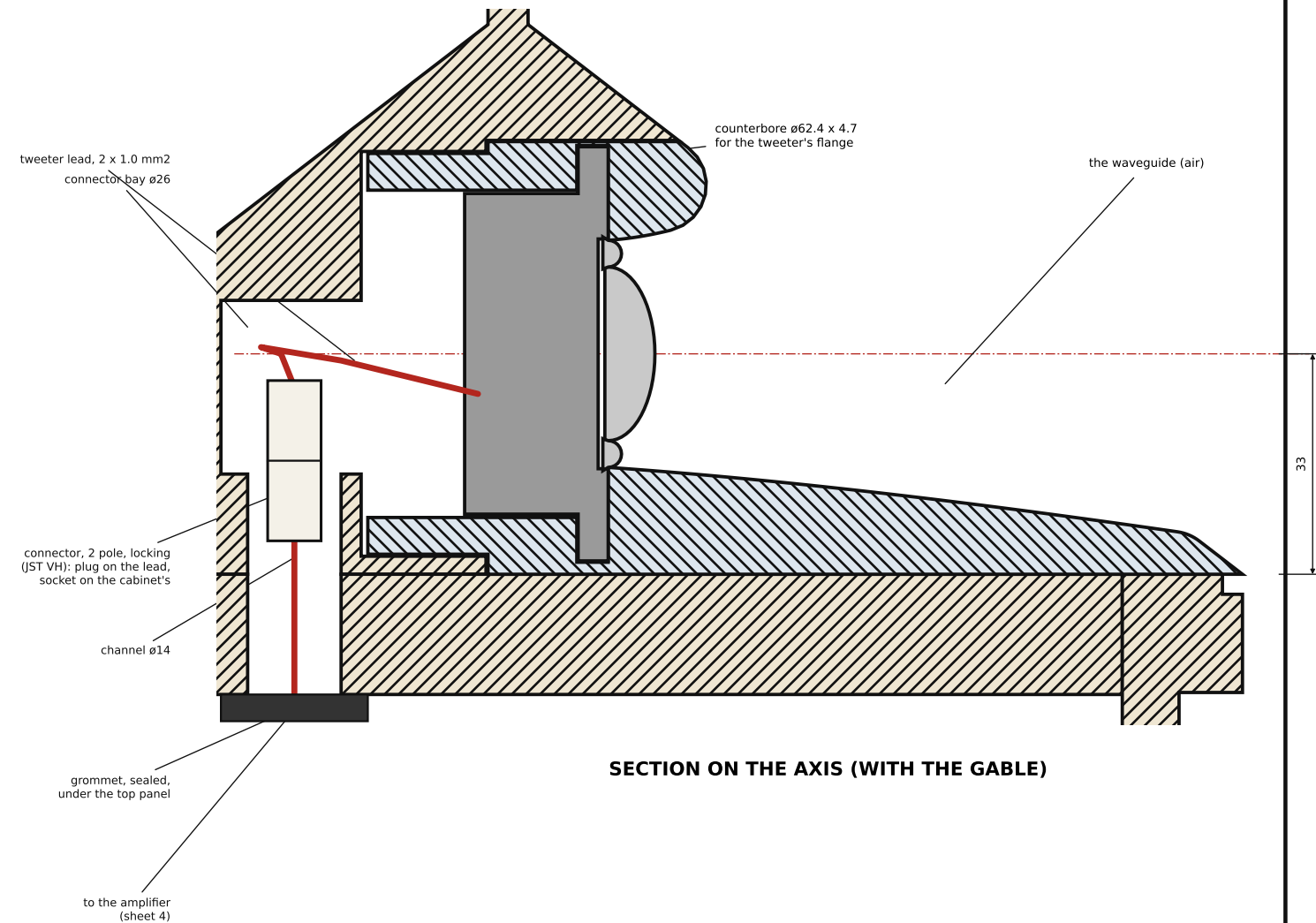
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FRONT



BACK (THE FACE THAT MEETS THE POCKET)



SECTION ON THE AXIS (WITH THE GABLE)

Fitting the tweeter and the insert

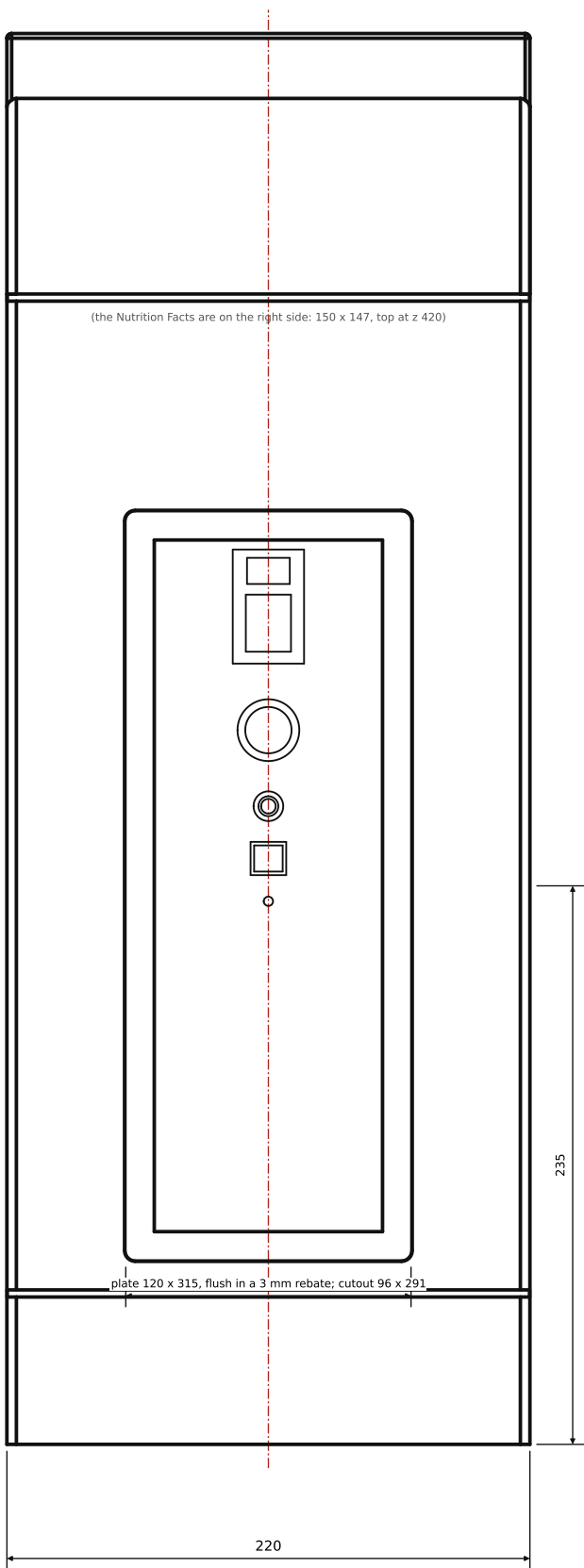
1. Solder the tweeter's lead (2 x 1.0 mm2, 120 mm) to its tabs and crimp the plug half of a two-pole locking connector (JST VH) on its end.
2. Seat the tweeter in the counterbore from behind, dome toward the throat, on a 0.5 mm foam gasket. Fix its 3 screws into the insert (heat-set brass inserts, M3).
3. Glue the magnets into the insert's back with epoxy, polarity marked, and their partners into the pocket's back wall, opposite poles out. Fit the two pins in the insert.
4. The cabinet's lead ends in the socket 100 mm above the grommet. With the insert out, draw the socket forward out of the bay through the empty pocket, plug the tweeter's lead into it, and tuck both and the slack back into the bay.
5. Slide the insert in, level, until the pins seat and the magnets pull it home: its face flush with the roof, the seam even.
6. Service: lift it out with a suction lifter on its flat border (or a pull loop under its lower edge), unplug, and the tweeter comes out with it.

earmilk bookshelf | Waveguide insert and tweeter mount

The insert at 1:1: front, section on the axis, back; the pocket; fitting and service.

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BACK

Mains
IEC C14 inlet, switch

Signal in
XLR (balanced) or RCA
USB for the DSP

Hypex FusionAmp FA122
2 channels, DSP crossover and EQ
125 + 125 W into 4 ohm
upright on the back
in its own sealed box (not airtight itself)
filters: Hypex Filter Design, over USB
mains in on the plate (IEC C14 with switch)

Tweeter, Scan-Speak D3004/602200
26 mm dome, faceplate behind the insert
1 mm2 pair, red +, black -

Woofer, SB Acoustics SB17NRX2C35-8
6 in, sealed box, DSP shelf to 45 Hz
1.5 mm2 pair, red +, black -

Runs, connectors and seals

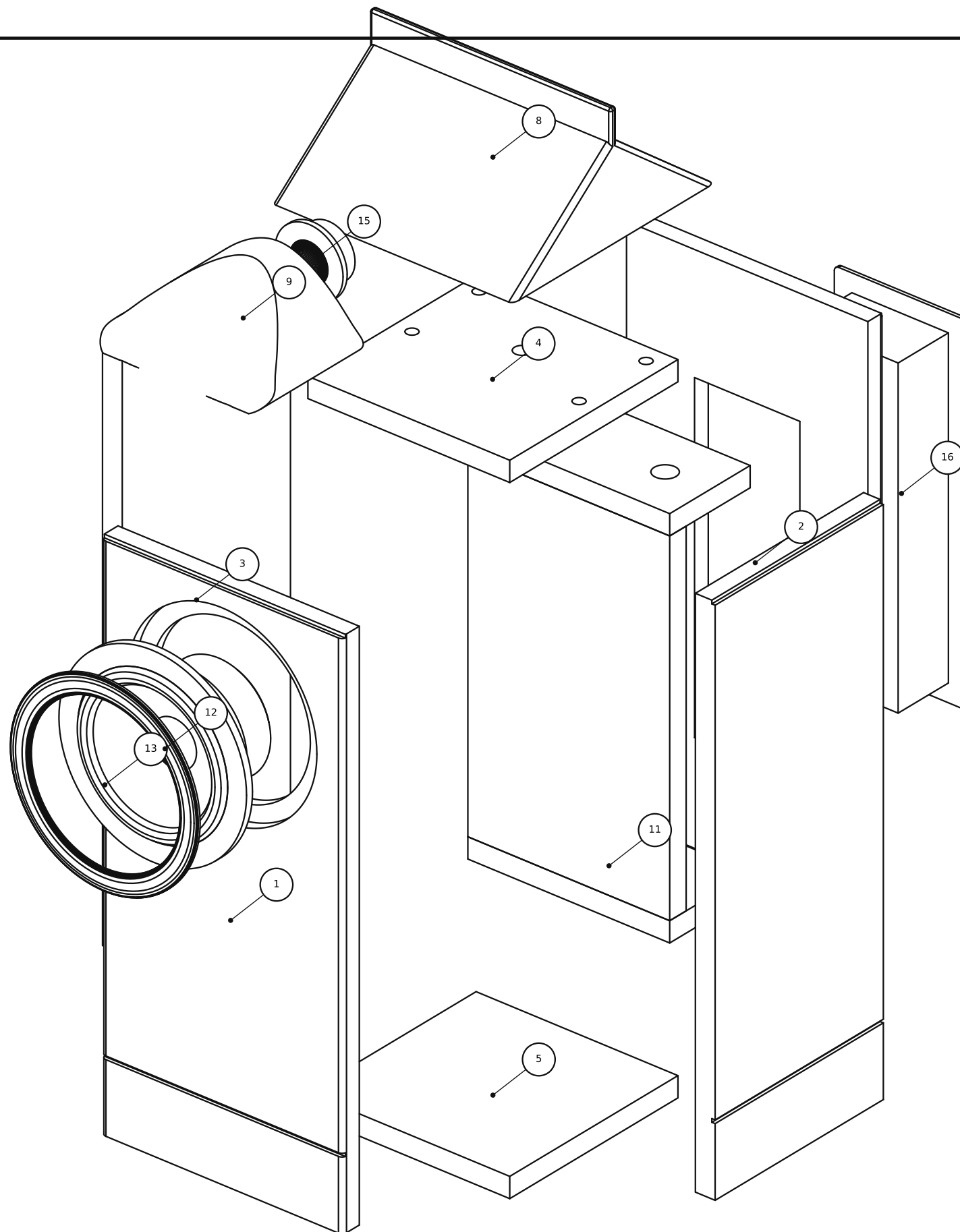
1. Woofer (CH1): from the amplifier box's gland to the woofer's terminals, 0.4 m of 1.5 mm2.
2. Tweeter (CH2): from the gland up through the brace's window to the top panel, through the 14 mm channel in the top panel and the gable block, 0.6 m of 1.0 mm2, ending in the socket of a two-pole locking connector (JST VH) in the bay behind the insert.
3. Seal the tweeter cable where it passes the top panel (grommet and sealant): the box is sealed, and the insert's pocket is open to the room through its seam.
4. The amplifier box's lid: a 16 mm gland for the two pairs; seal it.
5. The crossover (about 2 to 2.5 kHz), the woofer's shelf to 45 Hz (+7.6 dB) and the delays are set in the DSP. Wire red to + throughout.

earmilk bookshelf | Back and wiring

The back at 1:3 and the amplifier, drivers, cables and connectors. Active: the amplifier's DSP is the crossover.

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Parts

- 1 Front baffle x1**
18 mm Baltic birch, CNC: the woofer's cutout and rebate
- 2 Back x1**
18 mm birch: the amplifier's cutout and rebate
- 3 Sides x2**
18 mm birch; the Facts printed on the right one
- 4 Top panel x1**
18 mm birch, the tweeter cable's 14 mm hole, four dowel holes
- 5 Bottom panel x1**
18 mm birch
- 6 Window brace x1**
18 mm birch
- 8 Gable block x1**
laminated birch or printed whole: the insert's pocket and the cable channel
- 9 Waveguide insert x1**
printed (SLA, MJF) or CNC from solid; magnets, pins
- 11 Amplifier box x3**
18 mm birch floor, lid (gland) and front, glued and sealed
- 12 Woofer x1**
SB Acoustics SB17NRX2C35-8, 6 in coated paper
- 13 Trim ring x1**
printed, satin black or gunmetal, over the frame and its screws
- 15 Tweeter x1**
Scan-Speak Illuminator D3004/602200, faceplate behind the insert
- 16 Amplifier x1**
Hypex FusionAmp FA122, DSP, 2 channels

earmilk bookshelf | Exploded view and parts

The parts pulled apart along the way they go together, and what each one is.

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